Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Multi Observation Thematic Assembly Centre

Multi Observation Global Ocean Sea Surface Salinity and Sea Surface Density Product

MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013					
1.0	04/09/20	All	Initial version of the document Created from CMEMS-MOB-QUID-015-001 and CMEMS-MOB-QUID-015-002	B. Buongiorno Nardelli, A.Pisano	S. Guinehut
1.1	15/08/21	I.1	Minor changes to specify the update frequency and the temporal coverage of rep datasets	B. Buongiorno Nardelli	S. Guinehut
1.2	16/06/2022	All	Minor changes to specify the inclusion of SMAP observations	B. Buongiorno Nardelli, A. Pisano, M. Sammartino	S. Mulet
1.2	01/08/2022	All	Update of the validation statistics computed on the present version of the nrt and rep products (L4 v8.0). Update of the comparison against the previous version of the products (L4 v5.0)	B. Buongiorno Nardelli, A. Pisano, M. Sammartino	S.Mulet
1.3	16/06/2023	All	Major upgrade to provide an improved product at higher spatial and temporal resolution	B. Buongiorno Nardelli, A. Pisano, M. Sammartino	H.Etienne
1.4	23/10/2024	All	Updated description of processing algorithm to account for adjustment procedure applied to pre-satellite SSS period in MY datasets. Updated validation metrics including full MY period	M. Sammartino, B. Buongiorno Nardelli	H.Etienne

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I EXECUTIVE SUMMARY

I.1 Product covered by this document

This document describes the external product **MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013**, developed by the Centro Nazionale di Ricerca (CNR) Production Unit (MULTIOBS-CNR-ROMA-IT), which provides daily global gap-free Level-4 (L4) analyses of the Sea Surface Salinity (SSS) and Sea Surface Density (SSD) at 1/8° of spatial resolution. The product is obtained through a multi-dimensional (multivariate) optimal interpolation (OI) algorithm that combines sea surface salinity images from multiple satellite sources, such as NASA's Soil Moisture Active Passive (SMAP) and ESA's Soil Moisture Ocean Salinity (SMOS) satellites, with in situ salinity measurements and satellite Sea Surface Temperature (SST) information (Buongiorno Nardelli, 2012; 2016; Droghei et al. 2018). It includes 4 datasets:

- cmems_obs-mob_glo_phy-sss_nrt_multi_P1D, which provides near-real-time (NRT) daily data.
- cmems_obs-mob_glo_phy-sss_nrt_multi_P1M, which provides near-real-time (NRT) monthly data.
- cmems_obs-mob_glo_phy-sss_my_multi_P1D, which provides multi-year reprocessed (REP) daily data.
- cmems_obs-mob_glo_phy-sss_my_multi_P1M, which provides multi-year reprocessed (REP) monthly data.

REP time series is extended in time once a year. The temporal coverage of the REP dataset is from 1993 up to 12 to 24 months before present. The REP datasets benefit from the highest quality ocean observations inputs and are produced with a unique system (no change in time in the setup, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.

NRT datasets uses the NRT datasets of ocean observations inputs:

- INSITU_GLO_TS_ASSIM_NRT_OBSERVATIONS_013_047;
- SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001.

REP and NRT datasets are produced with the same system (same setup).

Note that these datasets supersede older datasets at lower spatial resolution (specifically: dataset-sss-ssd-nrt-weekly, dataset-sss-ssd-rep-weekly).

I.2 Summary of the results

MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013 quality is here estimated through comparison with fully independent in situ measurements collected by thermosalinographs (TSG) (CO_DMQCGL01_*_TS_TS_MERC.nc extracted from INSITU_GLO_TS_ASSIM_REP_OBSERVATIONS_013_051 (REP) datasets).

SSS (REP): The RMS difference with the thermosalinographs is equal to 0.33 psu in the open sea (>200 km offshore) and to 0.59 psu when including coastal data.

SSD (REP): The RMS difference with the thermosalinographs is equal to 0.32 kg/m^3 in the open sea (>200 km offshore) and to 0.52 kg/m^3 when including coastal data.

REP statistics have been compared with those computed from the benchmark dataset. No significant differences have been observed between the present HR dataset (L4 v1.1 HR) and the benchmark dataset (LR weekly L4 v8.0).

I.3 Estimated Accuracy Numbers

In the following, Table 1 provides the comparison between the accuracy estimate for the present HR SSS/SSD reprocessed product L4 v1.1 HR vs the previous LR weekly fields L4 v8.0 over the period 1993-2020.

Table 1: RMSD between reprocessed surface salinity and density fields and independent TSG measurements over the period 1993-2020 (95% confidence intervals were estimated through bootstrapping). The values in parentheses were obtained limiting the TSG matchups to data >200km offshore.

RMSD surface salinity (psu)		**RMSD surface density (kg/m^3)**	
cmems_obs-mob_glo_phy-sss_my_multi_P1D	dataset-sss-ssd-rep-weekly (benchmark)	cmems_obs-mob_glo_phy-sss_my_multi_P1D	dataset-sss-ssd-rep-weekly (benchmark)
0.59±0.01 (0.33±0.01)	0.58±0.01 (0.33±0.01)	0.52±0.01 (0.32±0.01)	0.51±0.01 (0.32±0.01)

II PRODUCTION SUBSYSTEM DESCRIPTION

II.1 MULTIOBS SSS/SSD processing

Table 2: MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013 product specification

Product Specification Customer Name	MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013
Geographical coverage	Global: 89.9375°S- 89.9375°N, 0.0625-359.9375°E
Variables	Sea Surface Salinity, Sea Surface Salinity Error, Sea Surface Density, Sea Surface Density Error and Sea Ice Fraction
Analysis	Yes
Forecast	No
Available time series	cmems_obs-mob_glo_phy-sss_nrt_multi_P1*: 01/01/2021 – ongoing cmems_obs-mob_glo_phy-sss_my_multi_P1*: 06/01/1993 to 12 to 24 months before present (see online catalogue)
Temporal resolution	Daily field Average Monthly field, nominally associated with the 15th of the month
NRT Target delivery time	Tuesday at 16:00
Delivery mechanism	Copernicus Marine Information Service (with a backup FTP)
Horizontal resolution	1/8° on a regular grid
Number of vertical levels	1 level (0 m depth)
Format	Netcdf 4.0 CF1.7

II.1.1 Short description

MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013 consists of global 2D sea surface salinity (SSS) and sea surface density (SSD) L4 fields, produced on a 1/8° regular grid at daily sampling. The NRT dataset (**cmems_obs-mob_glo_phy-sss_nrt_multi_P1D**) is produced weekly and made available each Tuesday (it provides the previous week´s daily analysis) and updated in delayed-time (each Tuesday it substitutes Near Real Time (NRT) from 4 weeks before with Delayed Time (DT) data). This configuration has been chosen to account for the delayed availability of SMOS data used in input. Differences in Estimated Accuracy Numbers between the NRT and DT data are negligible (well within the confidence interval provided). The cmems_obs-mob_glo_phy-sss_nrt_multi_P1M provides the monthly average of cmems_obs-mob_glo_phy-sss_nrt_multi_P1D (NRT monthly average is provided on the first of each month, and DT value is provided with 2 months delay).

The REP dataset (**cmems_obs-mob_glo_phy-sss_my_multi_P1D)** consists of a multi-year reprocessing of the global 2D sea surface salinity (SSS) and sea surface density (SSD) L4 fields (carried out on yearly basis), taking advantage of possible updates of upstream satellite and in situ data.

II.1.2 Input data sets

Four different data types are used by the new SSS/SSD processing:

1) The background (first guess) is taken from the previous day High Resolution (HR) analysis data in the NRT processing chain, and from a combination of the previous day High Resolution (HR) analysis data (or upsized Lower Resolution (LR) analyses if previous analysis is not present) over open-ocean and a new background over coastal areas. The new background is obtained by combining the upsized 1/4° SSS climatologies used for the previous product version and ad-hoc upsized Arctic climatologies taken from the Copernicus Marine Service platform (ARTIC_MULTIYEAR_PHY_002_003; https://doi.org/10.48670/moi-00007).

2) In situ data are taken from the surface values of the Conductivity, Temperature, and Depth, (CTD) profiler and Argo[1] float measurements extracted from two internal Copernicus Marine Service products providing Global Ocean- In-Situ temperature and salinity Near-Real-Time/Reprocessed Observations (INSITU_GLO_TS_ASSIM_NRT_OBSERVATIONS_013_047 (NRT) and INSITU_GLO_TS_ASSIM_REP_ OBSERVATIONS_013_051 (REP)). These data are quality controlled and homogenized profiles/time series Temperature/Salinity by concatenating the data per day and type to reduce the total number of data files and by subsampling the in situ profiles/time series in time and space to limit redundant information.

3) The L3OS 2Q (Level 3 Ocean Salinity corrected for seasonal latitudinal bias and coastal bias) de-biased daily valid ocean salinity values product from SMOS satellite. This dataset is produced and disseminated by the Centre Aval de Traitement des Données SMOS (CATDS, 2017). It is an intermediate product that provides, in daily files, SSS corrected from land-sea contamination and latitudinal bias, as detailed in Boutin et al. (2018). Ascending and descending orbits are processed separately by CATDS, but merged here after substantial subsampling based on a maximum estimated SSS error from model of 0.8 (see specifically the CATDS Algorithm Theoretical Basis Document, CATDS, 2017).

4) The SSS fields from NASA's Soil Moisture Active Passive (SMAP) satellite provided by the NOAA CoastWatch/OceanWatch. These data are 0.25◦ longitude/latitude Level-3 gridded sea-surface salinity (SSS) daily means, obtained by box averaging of SMAP satellite observations over the global ocean. Daily files are available back to the beginning of 2015 at https://coastwatch.noaa.gov/cw/satellite-data-products/sea-surface-salinity/smap.html. CoastWatch/OceanWatch Level-3 SSS products are generated, directly, from the NASA Jet Propulsion Laboratory (JPL) SMAP Level-2B SSS near-real-time swath HDF5 files, minimizing the data latency to within 24 h of Level-2 data availability (Fore et al., 2020; NASA/JPL 2020). As well as SMOS products, these data have, also, been used as an input to the interpolation (Sammartino et al., 2022).

5) The Copernicus Marine Service SST L4 near-real-time/multi-year reprocessed Operational Sea Surface Temperature and Sea Ice Analysis (OSTIA) developed by the U.K. Met Office (SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001; https://doi.org/10.48670/moi-00165; SST_GLO_SST_L4_REP_OBSERVATIONS_010_011; https://doi.org/10.48670/moi-00168).

[1] https://argo.ucsd.edu/ (last accessed 28/11/2023)

II.1.3 Method

This L4 product is obtained by interpolating in-situ and multi-sensor satellite-derived SSS and SSD data with the multidimensional OI technique originally introduced by Buongiorno Nardelli (2012). By using pre-determined satellite SST L4 data, this OI algorithm is able to extract information on the surface patterns and therefore increases the effective resolution of interpolated fields. The technique, originally developed to interpolate in situ SSS only, has been modified by Droghei et al. (2016) to provide dynamically consistent SSD fields, and by Buongiorno Nardelli et al. (2016) and Sammartino et al. (2022) to ingest also satellite observations from both Soil Moisture Ocean Salinity (SMOS) and Soil Moisture Active Passive (SMAP) platforms. Further developments have been carried out in the framework of Copernicus Marine Service, as documented in Droghei et al. (2018) and described hereafter. SSS and SSD L4 are obtained by running the OI processing separately on either SSS or SSD data.

The the desired values ($x_{analysis}$) at the interpolation grid point ($x_{analysis}$ in Equation 1) is obtained as a weighted sum of the anomalies of N observations ($y_{observed}$) with respect to the first guess represented by the background (x_{first_guess}) field. Differing from the previous version of this product, here, the background field is built by combining the upsized 1/4° SSS climatologies used in the previous product version and ad-hoc climatologies built from a Copernicus Marine Service product (ARTIC_MULTIYEAR_PHY_002_003) to account for the Arctic area. The weights provide the minimum expected estimate error (in a least square sense) and the estimate is unbiased (i.e., it has the same mean as the true field):

$$x_{analysis} = x_{first\ guess} + C(R + C)^{-1}(y_{obs} - x_{first\ guess}) \qquad \textit{(Eq. 1)}$$

where C represents the first-guess error covariance matrix, R represents the observation error covariance matrix (that is assumed diagonal, so that observation errors are uncorrelated and defined by constant values per each observation type/platform).

In the implementation considered here, the observation error covariance matrix is actually expressed as a noise-to-signal ratio (dividing it by signal variance). In the classic OI technique, the covariance function is generally estimated as a function of distance (thus defined in a bidimensional Euclidean space). However, depending on the input data available, covariance models are easily extended to multidimensional spaces (e.g., considering space–time variability) simply by introducing generalized distances in the covariance function. Here, the same multidimensional covariance model developed by Buongiorno Nardelli (2012) has been applied to interpolate either SSS or SSD measurements (Equation 1). From a purely theoretical point of view, this extended OI method can be interpreted as an approximation of a multivariate approach that includes both the SSS (or the SSD, depending on the processing considered) and the SST in the state vector used to build the observation matrix needed to estimate the covariance matrix in Equation 1. The cross terms in the covariance matrix are thus approximated by defining a covariance function that depends on both space–time distance and (high-pass filtered) thermal differences. In practice, the particular covariance model adopted gives more weight to the observations found on the same (high-pass filtered) isothermal of the interpolation point with respect to the observations found at the same spatial and temporal separation but characterized by different SST values. As the SST L4 data contain information at the mesoscale, this algorithm is thus able to increase the effective resolution of the SSS/ SSD L4. The used covariance function (Equation 2) is a simple Gaussian function of the form:

$$\boldsymbol{C}(\Delta \boldsymbol{r}, \Delta \boldsymbol{t}, \Delta \boldsymbol{SST}) = \boldsymbol{e}^{-\left(\frac{\Delta r}{L}\right)^2} \boldsymbol{e}^{-\left(\frac{\Delta t}{\tau}\right)^2} \boldsymbol{e}^{\left(\frac{\Delta SST}{T}\right)^2} \quad \textbf{(Eq. 2)}$$

where Δr, Δt, and ΔSST are the spatial, temporal, and thermal separations, respectively, while L, t, and T are the spatial, temporal, and thermal decorrelation terms, respectively. Here, the values used for the decorrelation scales and filtering are those defined in Buongiorno Nardelli (2012), namely *L* =500 km, *t* =7 days, and *T =2.75 K*. The SST L4 data were high-pass-filtered (cut-off at 1000km) to compute the OI weights.

Since in situ observations are very sparse and their number particularly low especially during the first decade covered by the REP (before the advent of Argo profilers), pseudo-observations have been used as additional input in the open ocean. Differing from the previous product version, these pseudo-observations have been extracted from the previous day HR analysis field (or the upsized LR analyses if previous analysis was not present) sampled every 8 pixels. In the coastal area (distance from the coast<200 km), pseudo-observations are taken from the climatological background. This strategy allows to provide a more homogeneous effective spatial resolution even in cases of prolonged absence of data within the interpolation input data search space/time radius and more accurate coastal retrievals.

The improvement obtained with this procedure has been evaluated by comparing different background/pseudo-observation configurations. The in situ observations noise-to-signal ratio is fixed at 0.05, while it is fixed to 1.0 for pseudo-observations. A very high noise-to-signal value is associated with SMOS and SMAP L3 data (=2.). This assumption is equivalent to saying that an average of 20 SMOS or SMAP observations is required to obtain a super-observation whose error would be comparable to the in situ measurement noise. This is coherent with the averaging generally carried out to reduce SMOS L2/L3 noise per se. In fact, what we adopted here can also be interpreted as a more advanced averaging based on a multidimensional weight that is able to filter this noise while increasing the effective resolution of the interpolated product. Additionally, to reduce the computation effort, either in situ or satellite input data are sampled within a searching radius of 1500 km, with a temporal interval of 15 days in the case of in situ measurements, and 3 days in the case of SMOS and SMAP observations, giving priority to the SMAP passages in the filling of the input matrix for the OI process (Sammartino et al. 2022).

The large scale interpolated HR fields have been adjusted to the LR data to avoid large scale biases in the MY products, induced by excessive persistence of the previous day background in the pre-satellite SSS period. These were noticed in the first version of the datasets released in November 2023.

III VALIDATION FRAMEWORK

MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013 validation was carried out by comparison with fully independent (i.e. not included in the processing) thermosalinograph (TSG) data to estimate the accuracy of the analysis (SSS/SSD-W-CLASS4-TSG-RMSD-GLOBAL in Table 4; see also Sections IV 1.1 and 1.3) and by computing spatial power spectral densities in selected geographical subdomains to evaluate the data effective resolution (SSS/SSD-W-CLASS1-PSD_XY-DOMAIN in Table 4; see also Section IV 1.2 and 1.4).

The TSG observations used for the validation are those included in the internal INSITU_GLO_TS_ASSIM_REP_OBSERVATIONS_013_051 product.

Table 3: Metrics used for the validation of MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013.

Metric name	Description
SSS-W-CLASS4-TSG-RMSD-GLOBAL	RMSD between TSG observations and L4 analysis for sea surface salinity
SSS-W-CLASS1-PSD_XY-DOMAIN With DOMAIN: • NATL • SATL • SPAC • IND	Spatial power spectral densities for SSS computed over selected sub-domains DOMAIN: • NATL: North Atlantic box: 50°W-25°W; 20-38°N • SATL: South Atlantic box: 5-30°W; 30-52°S • SPAC: South Pacific box: 110-138°W; 0-48°S • IND: Indian Ocean box: 60-90°E; 5-48°S.
SSD-W-CLASS4-TSG-RMSD-GLOBAL	RMSD between TSG observations and L4 analysis for sea surface density
SSD-W-CLASS1-PSD_XY-DOMAIN With DOMAIN: • NATL • SATL • SPAC • IND	Spatial power spectral densities for SSD computed over selected sub-domains DOMAIN: • NATL: North Atlantic box: 50°W-25°W; 20-38°N • SATL: South Atlantic box: 5-30°W; 30-52°S • SPAC: South Pacific box: 110-138°W; 0-48°S • IND: Indian Ocean box: 60-90°E; 5-48°S.

IV VALIDATION RESULTS

IV.1 Sea surface salinity and density

The TSG matchup database used for the multi-year reprocessed SSS/SSD (cmems_obs-mob_glo_phy-sss_my_multi_P1D) validation covers the period 1993-2020. It was built by remapping TSG data on the output grid of the HR dataset, selecting all measurements collected within ±12h with respect to the nominal interpolation date of the LR product and comparing them with the analysis value. The RMSD were computed both by considering all TSG data and limiting to offshore data (>200 km from coast), that are not affected by the strong salinity variance observed in the coastal areas close to the main river outflows.

For simplicity, spatial wavenumber spectra (Power Spectral Densities, PSD) have been estimated here concentrating only on land free portions of the domain and averaging over the entire year. PSD were computed taking latitudinal variations only, and averaging the results obtained at each longitude and at each time step to obtain a single spectrum. To reduce spectral leaking, a Blackman-Harris windowing function has been applied before computing the Fast Fourier Transform.

Four different geographical subdomains have been considered as described in Table 4:

- North Atlantic box: 50°W-25°W; 20-38°N
- South Atlantic box: 5-30°W; 30-52°S
- South Pacific box: 110-138°W; 0-48°S
- Indian Ocean box: 60-90°E; 5-48°S

The validation showed comparable results with respect to the LR version of SSS/SSD REP L4 v8.0 dataset, demonstrating a high variance even at lower wavenumbers.

IV.1.1 SSS-W-CLASS4-TSG-RMSD-GLOBAL and SSS-W-CLASS4-TSG-AVAIL-GLOBAL

cmems_obs-mob_glo_phy-sss_my_multi_P1D: Higher RMS error is found when considering reprocessed data over the entire domain (0.59±0.01) than in the offshore (0.33±0.01). Over time (figure 1), the RMSD computed from REP daily is comparable to the LR version of the dataset, with differences that always stayed within the confidence interval (estimated through bootstrapping as two standard deviations). From table 5 no significant differences are observed between the LR (L4 v8.0) and the present HR dataset (L4 v1.1 HR). The number of valid matchup data, however, is highly irregular along the time series, with matchups remaining mostly confined to the mid-low latitudes and prevalently located in the Northern Hemisphere.

Table 4: Validation statistics of REP SSS vs independent TSG data over the period 1993-2020 (offshore is defined as >200 km from coast).

dataset	cmems_obs-mob_glo_phy-sss_my_multi_P1D	dataset-sss-ssd-rep-weekly (benchmark)
# matchup	44858	44858
RMSD SSS	0.59±0.01	0.58±0.01
# matchup offshore	13354	13354
RMSD SSS offshore	0.33±0.01	0.33±0.01

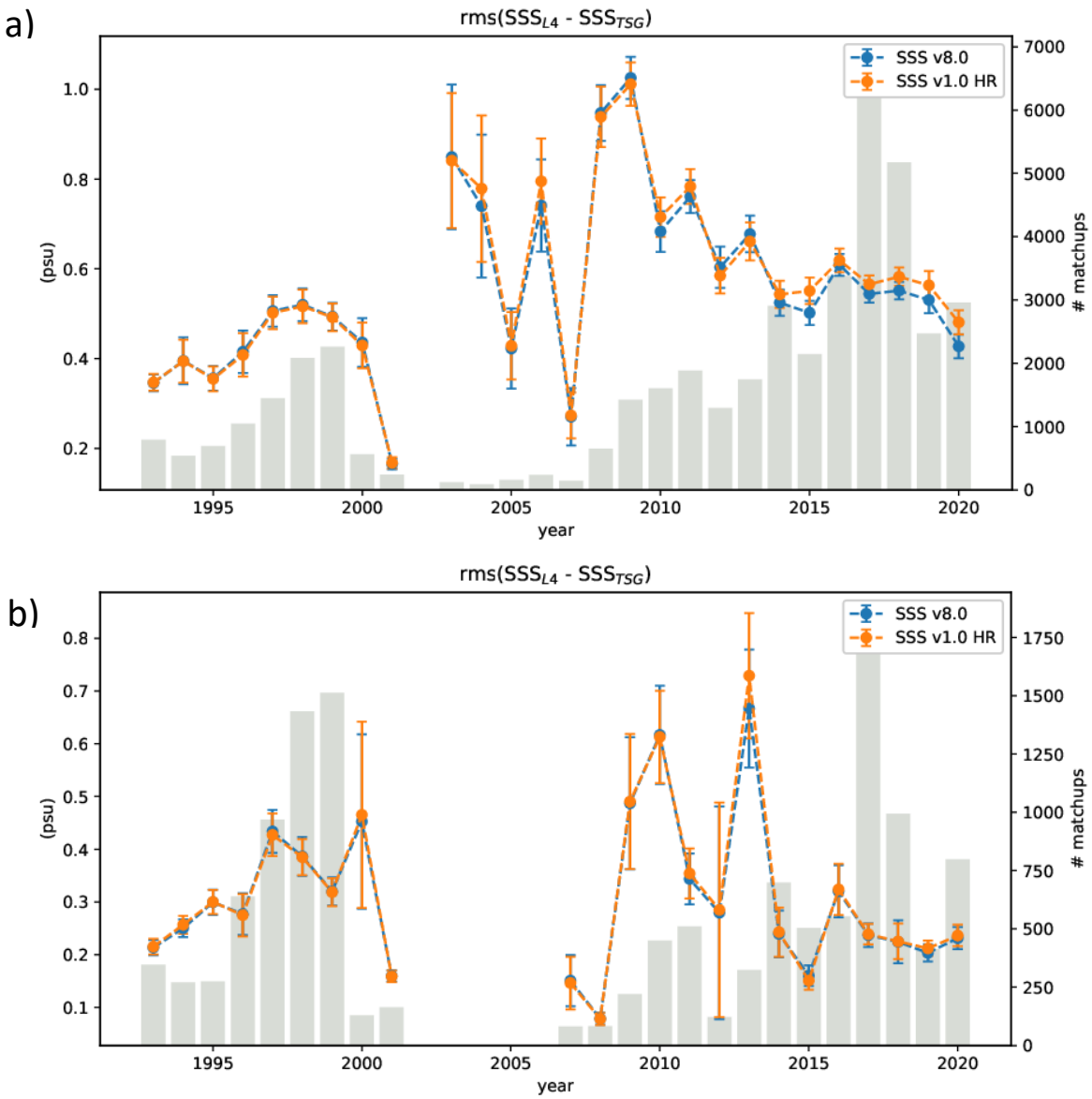

Figure 1: Time series of RMSD (and confidence interval) between TSG observations and present version of cmems_obs-mob_glo_phy-sss_my_multi_P1D (orange), and previous LR version (blue), for sea surface salinity over the period 1993-2020. The number of co-located observations available each year is shown by light grey bars in the background. The statistics were obtained using (a) all TSG data available (including coastal areas), (b) limiting to TSG data collected offshore (>200 km).

IV.1.2 SSS-W-CLASS1-PSD_XY-DOMAIN

As expected, the SSS-PSD computed from the background field shows an abrupt variance drop already at low wavenumbers (<0.2 deg^{-1}). Conversely, the new Copernicus Marine Service SSS L4 v1.1 HR spectra (irrespective of the dataset considered) show the highest variance even at lower wavenumbers, with an effective resolution of about eight times the grid resolution. At lower wavenumbers, noise finally flattens the spectrum (see also Droghei et al., 2018). The spectral behaviours are consistent among all subdomains.

Figure 2 gives the results for the 4 geographical subdomains considered, based on the cmems_obs-mob_glo_phy-sss_my_multi_P1D:

- SSS-W-CLASS1-PSD_XY-NATL
- SSS-W-CLASS1-PSD_XY-SATL
- SSS-W-CLASS1-PSD_XY-SPAC
- SSS-W-CLASS1-PSD_XY-IND

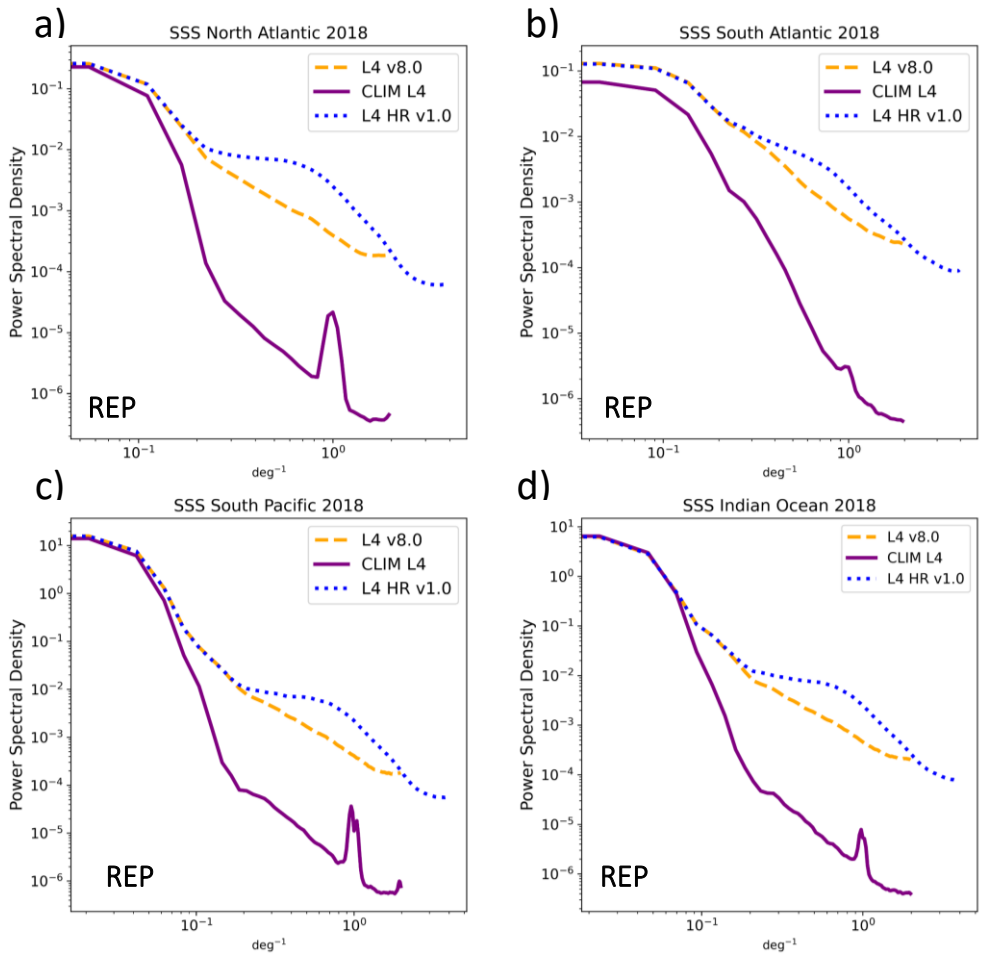

Figure 2: cmems_obs-mob_glo_phy-sss_my_multi_P1D SSS power spectral densities (PSD) computed for the four geographical subdomains: North Atlantic (a), South Atlantic (b), South Pacific (c), Indian Ocean (d) from previous weekly version of the dataset (orange dashed line), new version (blue dotted line), and climatology (purple solid line).

IV.1.3 SSD-W-CLASS4-TSG-RMSD-GLOBAL and SSD-W-CLASS4-TSG-AVAIL-GLOBAL

The same behaviours are found in the statistics computed from SSD data.

cmems_obs-mob_glo_phy-sss_my_multi_P1D: Higher RMS error is found when considering reprocessed data over the entire domain (0.52±0.01 kg/m^3) than in the offshore (0.32±0.01 kg/m^3). Over time (figure 3), the RMSD computed from REP daily is comparable to the previous weekly version of the dataset, with differences that always lay within the confidence interval (estimated through bootstrapping as two standard deviations). From table 6 no significant differences are observed between the LR (L4 v8.0) and the present HR dataset (L4 v1.1 HR). The number of valid matchup data, however, is highly irregular along the time series, with matchups remaining mostly confined to the mid-low latitudes and prevalently located in the Northern Hemisphere.

Table5: Validation statistics of REP SSD vs independent TSG data over the period 1993-2020 (offshore is defined as >200 km from coast).

dataset	cmems_obs-mob_glo_phy-sss_my_multi_P1D	dataset-sss-ssd-rep-weekly (benchmark version)
# matchup	44858	44858
RMSD SSD	0.52±0.01 kg/m^3	0.51±0.01 kg/m^3
# matchup offshore	13354	13354
RMSD SSD offshore	0.32±0.01 kg/m^3	0.32±0.01 kg/m^3

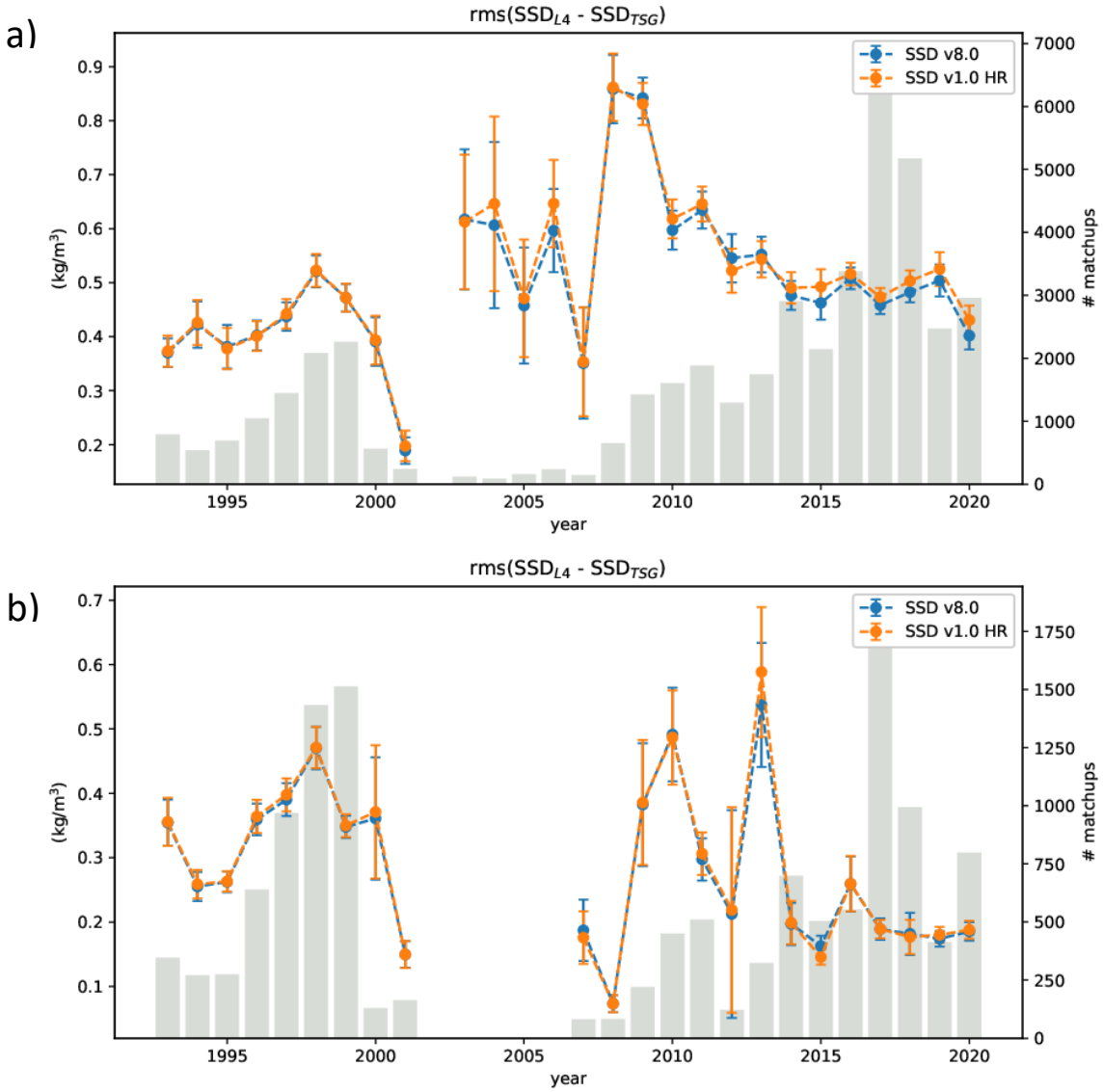

Figure 3: Time series of RMSD (and confidence interval) between TSG observations and present version of cmems_obs-mob_glo_phy-sss_my_multi_P1D (orange), and previous LR version (blue), for sea surface density over the period 1993-2020. The number of co-located observations available each year is shown by light grey bars in the background. The statistics were obtained using (a) all TSG data available (including coastal areas), (b) limiting to TSG data collected offshore (>200 km).

IV.1.4 SSD-W-CLASS1-PSD_XY-DOMAIN

The SSD PSD shows similar behaviours as SSS-PSD. The spectral behaviours are consistent among all geographical subdomains and demonstrate a higher effective resolution of the new version of the dataset with respect to the previous weekly one.

Figure 4 shows the results for the 4 geographical subdomains considered, based on the cmems_obs-mob_glo_phy-sss_my_multi_P1D:

- SSS-W-CLASS1-PSD_XY-NATL
- SSS-W-CLASS1-PSD_XY-SATL
- SSS-W-CLASS1-PSD_XY-SPAC
- SSS-W-CLASS1-PSD_XY-IND

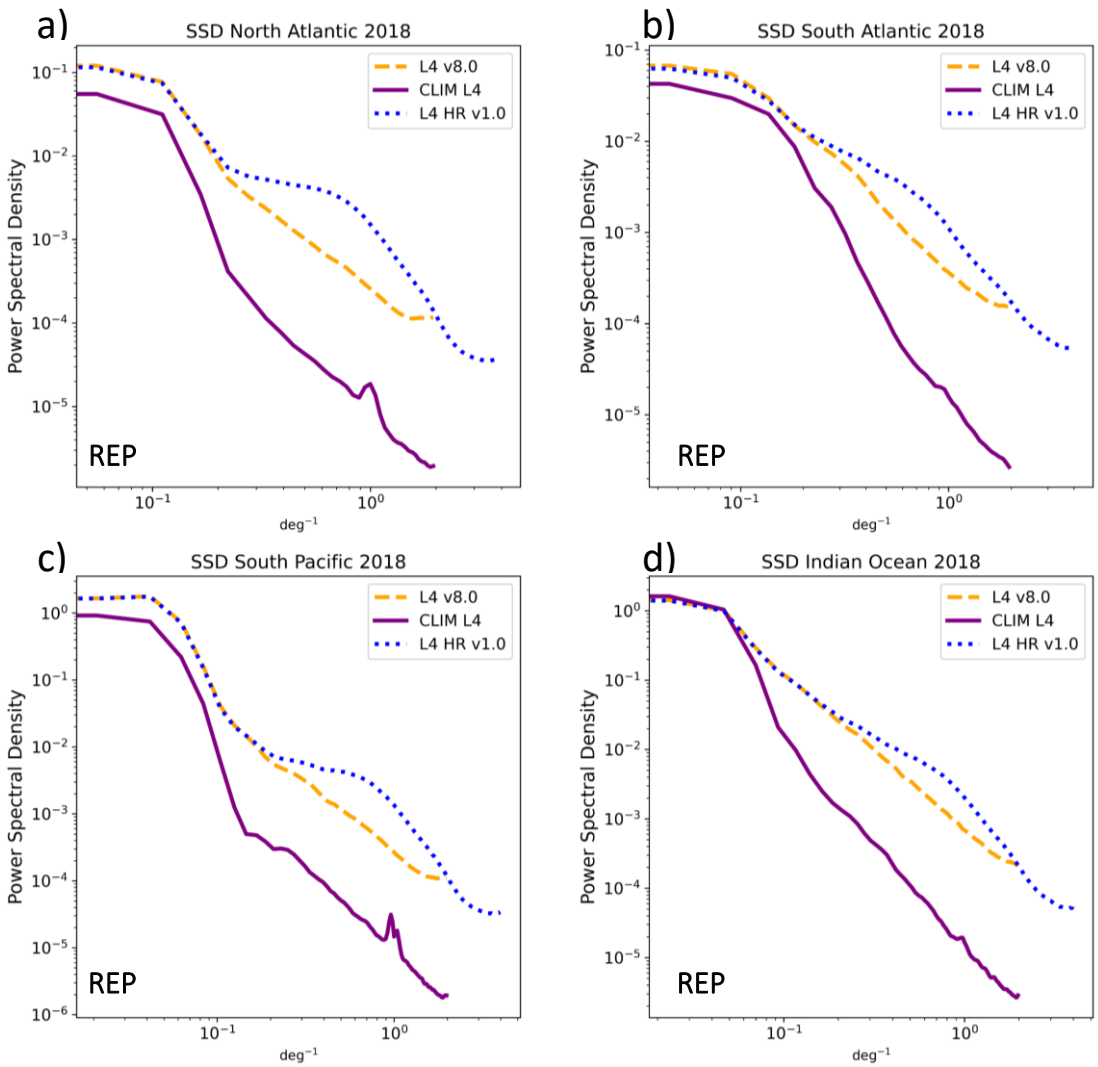

Figure 4: cmems_obs-mob_glo_phy-sss_my_multi_P1D SSD power spectral densities (PSD) computed for the four subdomains: North Atlantic (a), South Atlantic (b), South Pacific (c), Indian Ocean (d) from previous weekly version of the dataset (orange dashed line), new version (blue dotted line), and climatology (purple solid line).

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

There are no events to present with this version

VI QUALITY CHANGES SINCE PREVIOUS VERSION

VI.1 History of changes

Table 7: History of changes

December 2020	MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013 supersedes datasets (dataset-sss-ssd-nrt-weekly, dataset-sss-ssd-nrt-monthly, dataset-sss-ssd-rep-weekly, dataset-sss-ssd-rep-monthly) previously distributed as part of the following retired products: MULTIOBS_GLO_PHY_NRT_015_001 and MULTIOBS_GLO_PHY_REP_015_002. Upstream SST L4 data have been changed from NCEI (National Centers for Environmental Information, NOAA) to OSTIA (Operational Sea Surface Temperature and Sea Ice Analysis).
November 2022	The upstream SSS L3 data now include also NASA's Soil Moisture Active Passive (SMAP) satellite observations available from 2015 to present in addition to those of SMOS.
November 2023	A new SSS/SSD L4 daily product at 1/8° grid resolution is provided. This product includes a new mask and a background field obtained by combining the Global and ad-hoc Arctic climatologies.
November 2024	The MY SSS/SSD timeseries has been corrected for large-scale biases induced by excessive persistence of the background field in the pre-satellite SSS era.

VI.1.1 Changes in November 2023 release

Major upgrade of the processing chain to increase the SSS/SSD fields spatial and temporal resolution. The previous product is now superseded by a new SSS/SSD L4 daily field at 1/8° resolution.

VI.1.2 Changes in November 2024 release

A major correction to the pre-satellite SSS/SSD L4 data has been implemented to correct for large scale biases.

VI.2 Conclusion

The L4 v1.1 HR product quality demonstrated comparable statistics with respect to the LR version (L4 v8.0) over the full period as shown by comparing interpolated data with fully independent TSG data (see section IV). The PSD analysis demonstrated that the new product can resolve smaller spatial scales with respect to climatology and low resolution dataset.

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